

Inspiring healthy Food Choices in a Virtual Reality Supermarket by adding a tangible Dimension in the Form of an Augmented Virtuality Smartphone

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Figure 1: Supermarket in VR (left); Setup with smartphone and VR controller, the display is tracked via a RGB camera and fiducial markers (center); Usage of real smartphone app (here TrackSugAR [<https://apple.co/3cvsX3K>]) in the simulation with AV (right)

ABSTRACT

We want to understand the changing shopping behaviour, influenced by health-targeting nutrition apps on mobile devices. To achieve that, we have built a realistic Virtual Reality (VR) supermarket simulation and addressed core aspects such as handling virtual products, prevention of cybersickness, rounded off with an initial pilot study. On top of that, we built a virtual replica smartphone in VR with nutrition-related functionality. This has been extended with an Augmented Virtuality (AV) feature, that enables us to track the screen of a participant's own smartphone, hence allowing us to integrate real-world apps and letting the user interact with them during the simulation. To achieve a high-quality tracking, we propose a hybrid approach, utilizing an add-on RGB camera on the VR headset fused with data provided through a WLAN connection in the case of self-made apps. This enables the users to manipulate the simulation from within the smartphone app, introducing a versatile, highly usability-centered controller because they can handle their phone naturally.

Index Terms: Virtual Reality—Augmented Virtuality—Food Choices—HCI; Behavioral Change—Virtual Supermarket—

1 INTRODUCTION

In recent years, Virtual Reality (VR) has undergone a process of major development in the area of affordability, user-friendliness and thorough integration of technologies such as tracking-capable cameras. This makes it possible to involve information of the real world into VR, called Augmented Virtuality (AV). Virtual environments often lack a dedicated focus on usability and the integration of important tangible aspects. In this project, we designed a VR supermarket and integrated the usage of a physical smartphone. This way, the consumer can use real smartphone apps in VR. The goal is to investigate the influences of health-related nutrition apps on the shopping behaviour. Usability challenges were investigated, e.g. avoiding cybersickness or the readability of text. After hitting limitations of a virtual smartphone, we propose the process of integrating the real smartphone via AV. For people without experience in VR, the AV

smartphone can be an anchor point to overcome initial difficulties. Furthermore, immersion and presence are enhanced.

2 VR SUPERMARKET - SYSTEM DESIGN

In order to understand the consumer behavior, a virtual shopping environment for the VIVE VR HMD has been developed (see Figure 1). To counter missing sensorial input and to provide a realistic shopping experience, emphasis has been put on the immersion of the user. We followed the suggestions provided by Porcino et al. [3] to minimize cybersickness through the tunnel technique. The store layout can influence the customer behavior because it affects the shopping atmosphere. Stores are organized as departments (clustering products). In the industrial department approach [1] there are common layouts: Grid (rectangular, maximize floor space), race track (circular, focus on product categories) and free flow (mixed, focus on individual product categories). Often a mixture in layouts is present with a strong focus on a grid layout [1]. As we want to maximize realism, we plan to carefully model the specific layout for each use case in the configuration mode of the app.

2.1 Supermarket Configurator

We provide two modes in our Unity application, one to configure a supermarket layout and one for evaluating user experiences in the designed markets. In the configurator, different types of shelves can be placed and positioned as wanted. On the shelves, local product brands can be freely placed in any quantity and spacing necessary. Furthermore, a 3D object can be reskinned by adding a different label and it is possible to print text on it. It is important to add readable text, e.g. for nutrition tables.

2.2 Pilot Study - Survey with basic Questionnaire

In the VR mode, the seven participants all felt a sense of immersion. An important aspect of the evaluation was the readability of the replaceable textures on the products. The sticker on the test bottle had a resolution of 586x708 pixels and had text written on it in various font sizes. All participants who wore glasses were able to read the text down to a font size of 20, without glasses it was readable up to a size of 16. Three participants had no feeling of dizziness, while four chose "slightly agree" after playing 20 minutes. This can be explained by the achieved 40-50 frame rate on the outdated GPU

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(Nvidia GTX 970). This shows the need to address cybersickness even with the already involved counter measures.

3 THE SMARTPHONE - A DIVERSE SHOPPING COMPANION

We want to integrate this influential tool as a smart, user-operated interface into VR. It enables us to identify interfering and influencing factors in the decision-making process using smartphone apps.

3.1 Virtual Smartphone Foundation

As foundation a virtual campus resembling a smartphone was integrated. The interaction on the screen was implemented via the touchpad of the VR HMD controller. A (virtual) "pocket", in which the smartphone can be stored has been incorporated too. A globally working notification system is used to externally notify the user with tailored messages (e.g. instructions), independent of where they are currently operating in the VR world. A virtual camera feature with nutrition scanner is able to retrieve additional information, e.g. nutritional facts of any product in the taken image. In Unity smartphone sensor input can be virtually simulated to collect data during the evaluation phase, e.g. acceleration, orientation and position data (resembling their physical counterparts). As users cannot see their fingers in the virtual world, a mixture of light touches and stronger presses on the controller's touchpad was necessary for the interactions with the UI. Light touches indicate the location of the finger, stronger ones execute the intended action.

Because of the Covid-19 pandemic, a focus interview (contactless approach) has been conducted. All seven participants had prior experience with VR. We preferred experienced users, as they have an understanding about VR systems, hence can provide us with valuable input regarding usability and immersiveness. The realism of the VR smartphone was perceived by everyone as a realistic representation of a physical smartphone. Functionality such as the virtual "pocket" and haptic feedback through vibration for notifications were generally perceived as useful. The text input through an expandable keyboard was quite controversial. The keyboard input area needed to be extended over the virtual display canvas to be functional, but resulted in an unrealistic behaviour and thus hinders immersion.

3.2 Augmented Virtuality Smartphone

With the VR supermarket, we want to understand the purchasing decisions of the participants and enhance effectiveness of the development of health-oriented applications. With a solely virtual replica smartphone, it is possible to mimic similar behaviour of mobile applications, but there are limitations. For example, the intuitive handling of the smartphone is not possible and the controller replica touchscreen can overwhelm the user, especially as the own fingers can't be seen. Furthermore, there is too much diversity for the simulation considering today's smartphone operating systems and mimicking real apps is tedious with the limited given UI elements. To address those restraints, we want to propose an AV concept, thereby integrating the user's own smartphone.

Trackable fiducial markers are being placed on each of the 4 corners of the screen and with an RGB camera, we are "cutting out" the screen and its content, including the fingers of the user. The own phone acts as an anchor in the VR world, giving participants a familiar feeling into their hands as most people can blindly handle their phone. Besides the familiarity, involving real apps allows for an useful pipeline, when evaluating their effectiveness, usability, etc. with a high level of confidence. A versatile controller that is independent from the HMD hardware can not only react to choices being made in the supermarket, but also has the potential to influence the user in new ways through e.g. visual information [4].

In our approach, utilizing fiducial markers for the display tracking allows us to build on established approaches with a high level of robustness. Each marker in every corner provides its own pose, hence the rotation of the plane (equivalent to the rotation of the smartphone

in the real world) can be estimated fairly reliably. We utilize the C# ported ArUco library [2] from OpenCV for our prototype. The combination of a plain marker detector and OpenCV allows us to have control on each aspect of the pipeline. As foundation in Unity we chose the virtual smartphone asset and overlay the cropped camera texture. The virtual smartphone functionality is still available and it is possible to overlay the live camera feed with e.g. notifications. For self-made apps, we utilize a newly developed Unity plugin

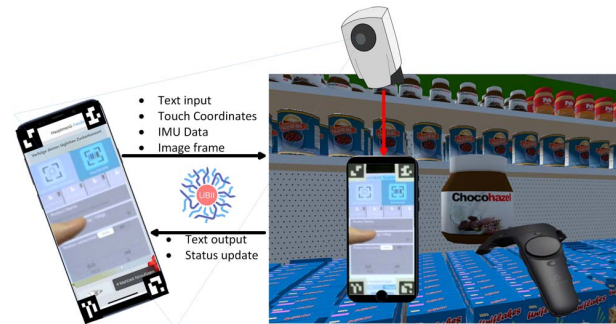


Figure 2: Contribution through Ubi-Interact

which provides the capability to connect the VR PC via WLAN with the smartphone (see Figure 2). The connection is being established through Ubi-Interact [5]. It can provide a stable transmission of e.g. text for input and output, the touch coordinates and smartphone IMU data. We could include visual cues when a touch occurs on the screen based on the coordinates or improve the smartphone rotation based on the IMU data to counter a jittery rotation when solely relying on the marker poses under difficult lighting conditions. A fusion of marker rotation vectors and transmitted IMU data could be especially robust (the marker pose does not suffer from drift). As Ubi-Interact is performance oriented, it is possible to stream the screen (e.g. through the generic Screen Capture API) and transmit the image in HD quality with 15-20 FPS.

4 CONCLUSION

In this work we describe solutions to usability challenges faced during the development of a VR supermarket. Furthermore, features for a virtual smartphone replica to interact with virtual nutrition products in VR have been explored. Our pilot study shows high potential. On top of that, we utilized an RGB camera and fiducial markers to integrate the user's physical smartphone and apps as overlay into the virtual world (AV). Lastly, we present a hybrid concept, where the camera input is extended with WLAN transmitted data to enhance the tracking and interaction with self-made applications.

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